



Bayesian Calculation Application of Medical Genetics

Note: This reference document contains the content of the Canvas page for “Bayesian Calculation Application of Medical Genetics” with minor modifications for presentation in a document format. Refer to this session’s Canvas page for the full interactive experience or if any hyperlinks do not function properly.

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Overview

A general problem in life and in healthcare is this: you have an estimate of the probability for something, but one with a lot of uncertainty associated with it. You feel that you can help the situation with an additional test, but this extra test still doesn't provide certainty, it just refines your estimate. However, how can this be integrated with prior knowledge? This is what we will try to answer in this session on Bayesian analysis. In order to keep this manageable, we will only apply this to genetic problems, but clinical testing and epidemiology are other places where Bayesian analysis could be applied.

Learning Objectives

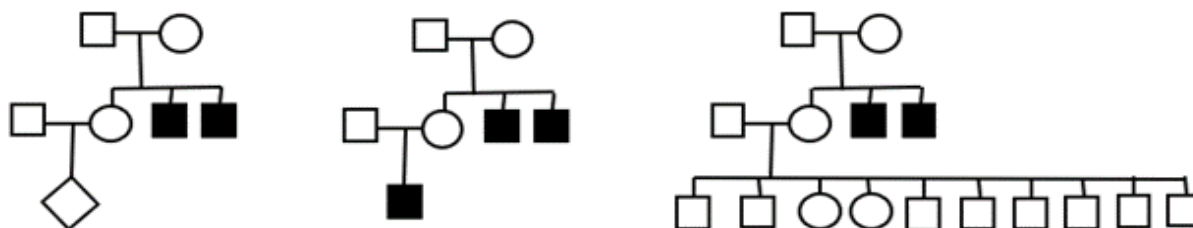
By the end of this session, you will be able to meet the following learning objectives:

1. Apply that Bayesian calculation is done when both of these two conditions are met:
 - Presence of doubt on carrier risk of a person, and
 - Presence of additional information influencing the risk for that person.
2. Demonstrate ability to include Bayesian calculation into risk calculation of genetic pedigrees including information from offspring phenotype, lab values or physical examination, or reduced penetrance.

Introduction

Bayes was a statistician, philosopher and priest living in the 18th century. The reason we still remember him is that he worked on statistics and developed what is known as the Bayesian theorem. One problem to which Bayesian analysis is applied is this: how does one take into consideration additional information after there is an initial estimate of how likely a certain event is?

To make you realize why this is important, let us think about the example of a person who could be a carrier of an X-linked recessive disease. In the left-most pedigree below, you will see a woman who is the sister of two people affected by hemophilia A. As you should remember from FFM1, Mendelian Risk Calculation, that means she has a carrier risk of $\frac{1}{2}$ and that therefore, her risk of having an affected child is $\frac{1}{8}$. But what happens if she has children?



Example of a person who could be a carrier of an X-linked recessive disease. Image Credit: NJ Larsen.

In the middle pedigree, this woman now has an affected child. What must you therefore conclude? She is therefore a carrier with 100% likelihood (your knowledge of her situation is improved; she cannot anymore be a non-carrier). That does not take much calculation, just application of logic.

However, the right-most pedigree shows an alternative outcome, where she has 10 healthy children: eight of these are boys and there are no affected children. In a purely intuitive way, you should say to yourself: the likelihood that a carrier would have 8 unaffected boys would be very low and that her probability of being a carrier therefore could not be very high. The next step is to try to quantify how low her risk of being a carrier is after having these children.

In the right-hand pedigree, just like in the left-hand pedigree, the woman starts with a 50% probability for being a carrier and 50% for not. (This will be defined as the **prior probability**, the probability that exists before taking additional information into account). Let us look at these two situations individually.

IF the woman is a carrier, then for each son she is having, there is a $\frac{1}{2}$ probability that offspring will be affected. For a carrier woman to have two healthy sons, the probability would be $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$. For eight sons, the probability would be $(\frac{1}{2})^8 = \frac{1}{256}$. If a carrier woman has a daughter, and her husband as expected is healthy, then the daughter has a very large probability of being healthy; this is true irrespective of whether the daughter received the disease-causing allele from the mother or not, because the father must have given a normal allele. The health of the daughters therefore is not informative, and we will ignore those. Notice, this paragraph starts with an IF, so it is actually a **conditional probability** we calculated: if the woman is a carrier then the probability of having 8 healthy sons is $\frac{1}{256}$.

On the other hand, if the woman is not a carrier, then the question is: why would any offspring be anything but healthy? Only through a new mutation would that appear, and the risk of new mutations would be in the range of 1/10,000 to 1/100,000. This risk is so low that we will ignore it. This is also a **conditional probability**: if the woman is not a carrier, then her probability of having 8 healthy sons is 1 (= 100%)



With these arguments, we are now ready to start calculating an adjusted risk for the woman to be a carrier. To do that we will use a table.

Arguments	Woman is a carrier	Woman is not a carrier	Sum
Prior probability	$\frac{1}{2}$	$\frac{1}{2}$	
Conditional = “Observation” – we observe 8 healthy sons	$\left(\frac{1}{2}\right)^8 = \frac{1}{256}$	1	
“Product” = joint probability = product of cells above	$\frac{1}{2} \times \frac{1}{256} = \frac{1}{512}$	$\frac{1}{2} \times 1 = \frac{1}{2} = \frac{256}{512}$	Sum of this row: $\frac{1}{512} + \frac{256}{512} = \frac{257}{512}$
Posterior probability = joint probability / sum	$\left(\frac{1}{512}\right) \div \left(\frac{257}{512}\right) = \frac{1}{257}$	$\left(\frac{256}{512}\right) \div \left(\frac{257}{512}\right) = \frac{256}{257}$	

The two first rows are presenting the calculations we made in the previous paragraphs in a different way. Prior probability is the $\frac{1}{2}$ for being a carrier and for being not-a-carrier. Notice that this row must sum to one, because there are only two possibilities (she is either one or the other of the two). Second row presents the conditional probabilities: notice that we calculated one value under the assumption of the woman being a carrier which is placed in the second column, and a different value under the assumption of the woman being not-a-carrier which is placed in column 3. Notice also my unofficial name for this row is “observation”, because it pays to write exactly the observation which forms the basis for the value entered. This row does not sum to one, because the content are not really probabilities.

The next row is officially called joint probability but really is a description of odds. However, my unofficial name for this is product, because the value entered is the product of the cells in the column above. However, it is important to notice that the content in this row are odds: value for “carrier” column is $\frac{1}{512}$ and for the “non carrier” column is



$\frac{1}{2}$. Those two numbers do not sum to one, and because they do not, they are not probabilities. However, if you make the two fractions have the same denominator (so $\frac{1}{2} = \frac{256}{512}$), it is pretty easy to see that the odds for being carrier versus non carrier is 1:256.

To change the odds to probabilities, we need to make a row in which the sum becomes one, and which still has the same odds for being one or the other. This process starts with summing the two values in the product row. However, as you may remember from prior training, before you can sum fractions, you need to have the same denominator, which is one reason $\frac{1}{2}$ became $\frac{256}{512}$.

The posterior probability is produced by taking the value in the product cell and divide by the sum. This is easier than it looks. The mathematically correct version is that when dividing by a fraction, you multiply with the inverse. When the two fractions have the same denominator, the denominator will thereafter cancel out. The shortcut is therefore, once all numbers have the same denominator, you remove the denominator and it becomes two regular divisions ($\frac{1}{257}$ and $\frac{256}{257}$). Because these two numbers sum to one, the posterior probability is a proper probability. The posterior probability is the new probability for the title of the column! In this case, that means that the probability for the woman being a carrier changed from $\frac{1}{2}$ (which was the probability when she was born) to ($\frac{1}{257} = 0.39\%$) after she had her 10 kids. Notice that Bayesian analysis cannot prove or disprove anything, meaning that the risk does not reach zero (but in this case pretty close).

Parentetical paragraph: another way to think about the last paragraph is the return to an earlier statement: the odds presented in the product line was 1:256. When you see this, you can also say: there are 1 + 256 total elements, = 257. The probabilities therefore become $\frac{1}{257}$ and $\frac{256}{257}$. Same result with slightly different argument.

See canvas for an interactive activity to learn more.

Prerequisites for Making a Bayesian Analysis

One important aspect of what is included in the table is the prior probability. For it to make any sense to make a Bayesian analysis, the numbers in this row have to be between 0 and 1, but not exactly 0 or 1. In other words, a prerequisite for making a Bayesian analysis is that there has to be doubt about the carrier risk; if there is certainty, then no reason to make the calculation. The second prerequisite for doing a Bayesian analysis is that there has to be additional information. Therefore, to reiterate:



Conditions for Bayesian analysis include:

- Doubt about the starting point (carrier risk)
- Presence of additional information

In the above example, the doubt existed: the woman had a prior probability of 50% for being a carrier and 50% for not being a carrier; and the additional information existed in the form of her 8 healthy boys.

Check your knowledge by completing the following exercise:

Knowledge Check Question

Question: Could you make a Bayesian analysis for carrier risk of an X-linked disease in a person with karyotype 46,XY.

Answer: For Hemophilia A the answer is NO, because either the person has an X with a healthy gene and is healthy, or the person has a disease-causing gene and shows the disease. The person who is healthy therefore cannot be a carrier of a disease-causing gene: there is no doubt! The possible exception is if there were an X-linked disease which shows reduced penetrance in a 46,XY person, such that a carrier could be healthy.

Other Situations in Genetics that Would be Amenable to Bayesian Analysis

There are a number of other situations where a Bayesian approach would be warranted. These generally differ from the above example in that the additional information is about health of the person him- or herself, while in the above example, additional information consisted of health of offspring.

The next example is a 38-year-old person who has a parent with an autosomal dominant disease which shows adult onset (the offspring in the below pedigree). He could be lucky by not inheriting the disease gene, or he could be a carrier who in a few years will start showing symptoms. An additional complication: not everyone carrying the disease allele will have onset at the same age. As an example, Huntington disease is caused by an expanded trinucleotide repeat, and the exact nature of that expansion is one of the factors influencing age of onset. This disease eventually causes all carriers to become affected = penetrance of 100% (You will hear more about this disease in another session during the reproductive module). Therefore, for those carrying a disease-causing allele, the cumulative probability of being affected by symptoms is often approximated with an S-shaped curve like the one below. The exact shape and position relative to the age axis varies from study to study, so do not memorize this exact curve.

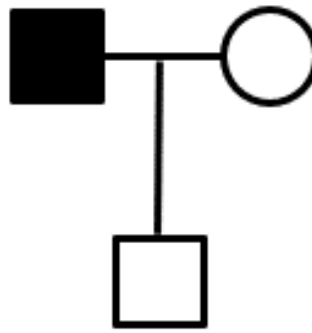


Image Credit: NJ Larsen.

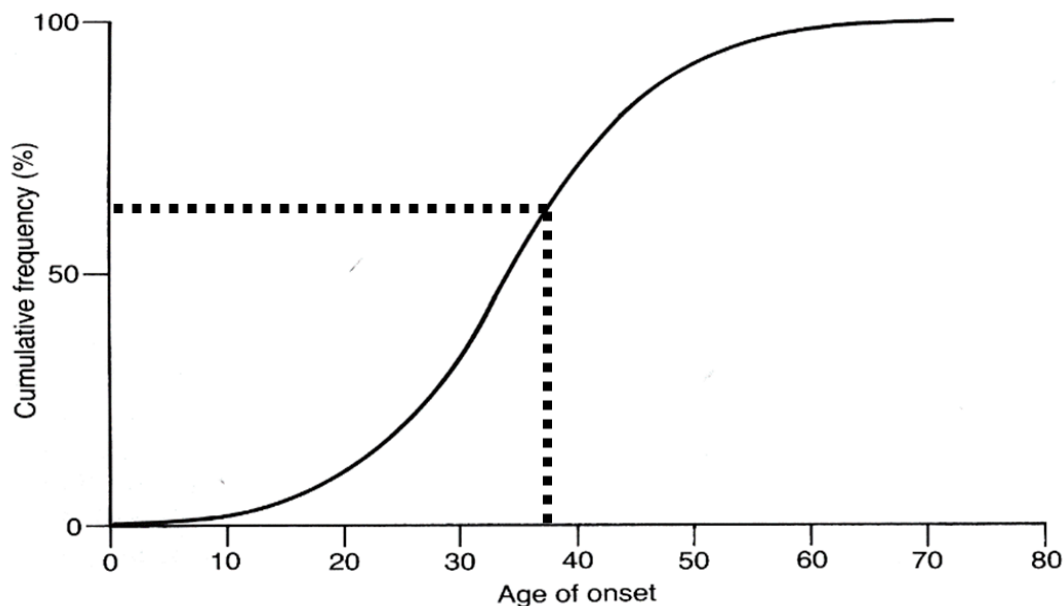


Image Credit: NJ Larsen.

To reiterate: for a person at an intermediate age, this person could be a carrier who has not yet started showing symptoms, or the person could be a “not a carrier”. It should be obvious from the above curve that the older the person is without having any symptoms, the lower the probability that the person is a carrier.

The stippled lines show how to read the graph for a person who is 38 years-of-age. Go from 38 on the X-axis to the curve and from that intersection to the Y-axis to determine that about 66% of carriers would already be showing symptoms at that age.

Now let us start building the Bayes table for this example. The prior probability is $\frac{1}{2}$ for being a carrier and $\frac{1}{2}$ for not. This is because the affected parent has an enormously larger probability of being a heterozygote than for being a homozygote. The offspring of a heterozygote has $\frac{1}{2}$ probability for having inherited the disease-causing allele (all of this is exactly what you should know from Risk Calculation in FFM1).



The observation is that the person is healthy at age 38. In a previous paragraph, we determined a probability of about 66% ($= \frac{2}{3}$) for a carrier to already show symptoms at that age. But you need to realize that the $\frac{2}{3}$ is the likelihood of being affected at that age; the person we are talking about is the opposite, not affected. The number we need is therefore $1 - \frac{2}{3} = \frac{1}{3}$ which is the likelihood that a carrier would be not affected at age 38.

See the second version of the figure below.

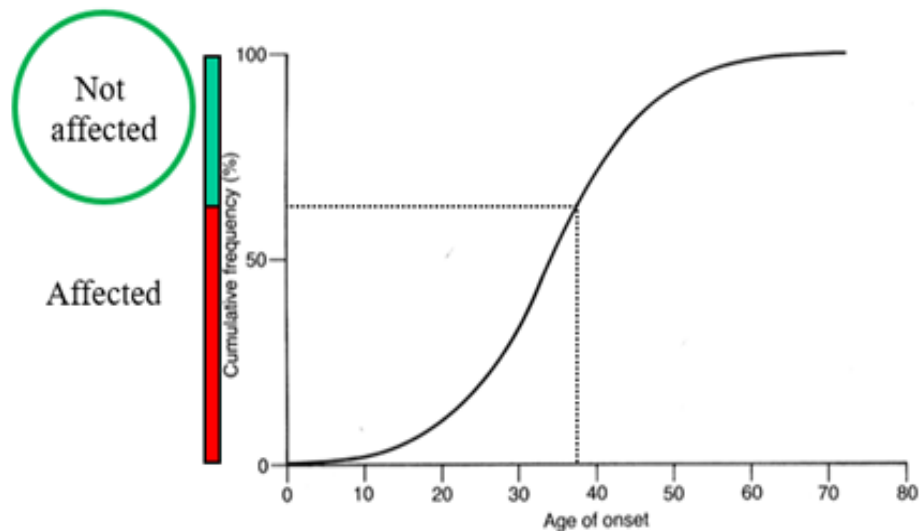


Image Credit: NJ Larsen.

On the other hand, if the person is not a carrier of a disease-causing gene, then the expectation is that the person would be healthy, so the second row becomes $\frac{1}{3}$ in the carrier column and 1 in non-carrier column. This is therefore the time to show the whole table; rows 3 and 4 are produced as in the first example.



Argument	Person is a Carrier	Person is not a Carrier	Sum
Prior probability	$\frac{1}{2}$	$\frac{1}{2}$	
Conditional = "Observation: II-1 is healthy"	$\frac{1}{3}$	1	
"Product" = joint probability	$\frac{1}{6}$	$\frac{1}{2} = \frac{3}{6}$	$\frac{4}{6}$
Posterior probability =	$\frac{1}{4}$	$\frac{3}{4}$	

In other words, the final result is that the carrier risk for the son in this example is $\frac{1}{4}$ at this point of his life.

Disease with Reduced Penetrance

In the pedigree below, person I-2 has an inherited disease, retinoblastoma. This is inherited as an autosomal dominant condition, but not everyone carrying the disease allele will show the condition. The exact penetrance of this condition is still up for debate, but for use at this point, we will assume that the penetrance is 80% for this family, equal to an 80% probability of showing the disease if the allele is present.

In the family, there is also person II-1. This person is healthy, does not show retinoblastoma. So either the person did not receive the disease causing allele or did receive but without showing the disease. What is the probability II-1 is a carrier for the disease-causing allele?

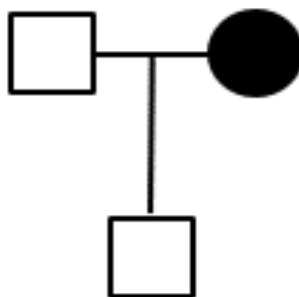


Image Credit: NJ Larsen.



As soon as you saw autosomal dominant inheritance, you should assume that the mother is a heterozygote and that the prior probability for the offspring to inherit the allele is $\frac{1}{2}$ (just as in the previous example). That means you know the first row in the table already. But stop and think for a second: what is the observation and how do you apply that in the second row?

The observation is that II-1 is healthy. The basis for interpreting this is the 80% penetrance, but remember that means 80% for being affected, the opposite of what you observe. Therefore, you need to utilize $1 - 80\% = 20\% (= \frac{1}{5})$ in the table under the heading of carrier: a carrier has a 20% probability of being healthy when the penetrance is 80%. For the column of II-1 being not a carrier, then he should be healthy, and you enter a "1". That produces the table seen below, where rows three and four is filled out as before:

Argument	Person is a Carrier	Person is not a Carrier	Sum
Prior probability	$\frac{1}{2}$	$\frac{1}{2}$	
Conditional = "Observation: II-1 is healthy"	$\frac{1}{5}$	1	
"Product" = joint probability	$\frac{1}{10}$	$\frac{1}{2} = \frac{5}{10}$	$\frac{6}{10}$
Posterior probability =	$\frac{1}{6}$	$\frac{5}{6}$	

See Canvas to review an interactive to learn more about Huntington Disease.

What if there is more than One Observation?

In some cases, there could be two different things you would need to take into consideration. This could for example be health of offspring as one observation and a test result as the second observation.

How could both be taken into consideration? Obviously, you could make two full Bayesian tables, where the result of the first table becomes the prior of the second table. However, it is simpler to have two rows for observations, and in the product row, multiply everything that is above that row.



Application Outside of Genetics

Consider the following text which is worded as a question:

A diagnostic test has a sensitivity of 99% and a specificity of 98% of detecting a specific disease. This test is introduced as a screening test into a population in Michigan. In this population, the prevalence of the disease of one in a thousand. A 38-year-old man from this population is screened with this test and the test result is positive indicating he might have the disease. What is the most likely interpretation of the result?

- The test is positive, he has the disease.
- The likelihood he has the disease is less than 5%.
- The risk is 98% that he has the disease.
- His risk of having the disease is between 25% and 35%.
- The prior probability always equals the posterior probability.

You may remember from epidemiology that the name for what is asked here is positive predictive value. One way to calculate the positive predictive value is to use a Bayesian approach. So first, what is the prior probability? That comes from the prevalence: there is a $\frac{1}{1000}$ (= 0.001) probability that he is affected. The second column would be that he is truly not affected, and the prior probability there would be $1 - \frac{1}{1000} = \frac{999}{1000}$. However, this is close enough to 1 that we will round it to 1.

The next row is observation: we observe a positive test result. The probability that a truly affected person gets a positive test result is given by the sensitivity (99%) (Sensitivity (true positive rate) is the probability of a positive test result when the individual truly is positive). The probability that a person truly not affected would get a positive test result is $1 - \text{specificity} = 1 - 98\% = 2\%$ (Specificity (true negative rate) is the probability of a negative test result, conditional on the individual truly being negative). The Bayes table therefore becomes:

Argument	Person is Affected	Person is not Affected	Sum
Prior probability	0.001	1	
Conditional = "Observation: test result is positive"	99%	2%	
"Product" = joint probability	0.00099	0.02	0.02099
Posterior probability =	0.047 = 4.7%	.953 = 95.3%	



Therefore, answer B turned out to be correct. In epidemiology, you learned that positive predictive value depends on the prevalence of the disease involved, and this clearly is demonstrated here. If you repeat the calculations with a prevalence of 1/100, the positive predictive value will be much higher.

Of course, you could also have used [an online calculator](#) of positive predictive value to make the calculation.

Another Application Outside of Genetics

Another situation, also not related to genetics, where a physician would need to use a similar approach is where a patient presents with symptoms that makes the physician suspect a severe condition while being aware that a more common, less severe cause is possible. Using a screening test at that point will not give a 100% answer to whether the person has the severe condition, but by using a Bayesian approach to combine information, the risk estimates would become more precise after use of the screening test. [ScholarRx has a brick](#) based on such a situation that you are welcome to check out, but please be aware that I am not going to test based on the brick. Also be aware that the mathematics is presented in a different way in that brick – so if you do not understand my presentation, maybe you will understand that way of presenting it.

A second approach of applying a Bayes argumentation to screening tests is found in this very nice video (It starts out a bit mathematical but ends up with a simplified approach.):

- [The Medical Test Paradox, and Redesigning Bayes' Rule](#) (21:13)